ELG3136 Lab 3 Pre-lab: Step-by-Step Solution

Frequency Response of a Common-Emitter BJT Amplifier

Given. $V_{CC} = 15$ V, $R_1 = 22$ kΩ, $R_2 = 12$ kΩ, $R_C = 3.3$ kΩ, $R_E = 4.3$ kΩ, $R_L = 4.7$ MΩ, $C_1 = C_2 = C_E = 10$ μF (C_1 input coupling, C_2 output coupling, C_E emitter bypass).

Assumed. 2N3904 with $\beta = 150$ (replace with your measured β), $V_{BE} = 0.7$ V, $V_T = 26$ mV, $V_A \approx 74$ V, $r_b \approx 10$ Ω; ideal signal source ($R_s = 0$).

Part A: DC operating point

Every task needs the bias point, because g_m, r_π, r_o, C_π and C_μ all depend on it.

Step A1: DC equivalent circuit. At DC all capacitors are open, so v_{in} and R_L drop out. Replace the divider R_1, R_2 by its Thevenin equivalent (Figure 1):

$$V_{Th} = V_{CC}\frac{R_2}{R_1 + R_2} = 15 \cdot \frac{12}{34} = 5.29 \text{ V}, \quad R_B = \frac{R_1 R_2}{R_1 + R_2} = 7.76 \text{ kΩ}$$

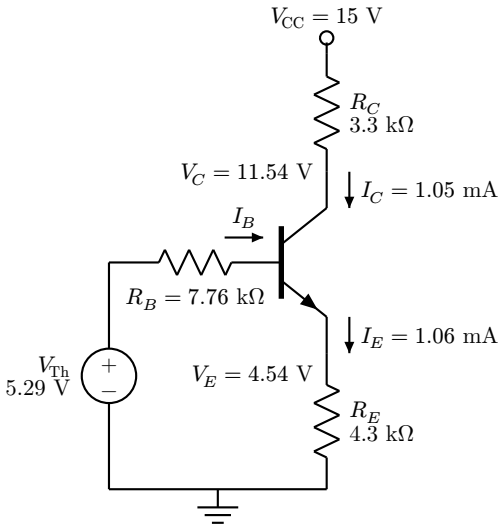


Figure 1: DC equivalent circuit with the Thevenin bias source. Currents and node voltages are the results of Steps A2 and A3.

Step A2: KVL around the base-emitter loop ($V_{Th} \to R_B \to V_{BE} \to R_E \to$ ground).

$$V_{Th} = I_B R_B + V_{BE} + (\beta + 1) I_B R_E \quad \Rightarrow \quad I_B = \frac{5.29 - 0.7}{7.76 \text{ k} + 151 \cdot 4.3 \text{ k}} = 6.99 \text{ μA}$$

Step A3: Currents and node voltages.

$$I_C = \beta I_B = 1.05 \text{ mA}, \quad I_E = (\beta + 1) I_B = 1.06 \text{ mA}$$

$$V_C = V_{CC} - I_C R_C = 11.54 \text{ V}, \quad V_E = I_E R_E = 4.54 \text{ V}, \quad V_B = V_E + V_{BE} = 5.24 \text{ V}$$

Step A4: Check the operating region. $V_{CE} = 7.0 \text{ V} > V_{CE,sat}$ and $V_{CB} = 6.3 \text{ V} > 0$, so the transistor is in forward active. I_C barely depends on β (1.04 mA for $\beta = 100$).

Step A5: Small-signal parameters.

$$g_m = \frac{I_C}{V_T} = 40.3 \text{ mS}, \quad r_\pi = \frac{\beta}{g_m} = 3.72 \text{ kΩ}, \quad r_o = \frac{V_A}{I_C} = 70.6 \text{ kΩ}, \quad R_L' = R_C \parallel R_L \parallel r_o = 3.15 \text{ kΩ}$$

Task 1: Add the BJT parasitic capacitances

Step 1.1: Identify them. The high-frequency hybrid-π model adds C_π between base and emitter (B-E diffusion plus junction capacitance) and C_μ between base and collector (reverse-biased B-C junction). A discrete transistor has no collector-substrate capacitance.

Step 1.2: Place them in the circuit (Figure 2, red dashed).

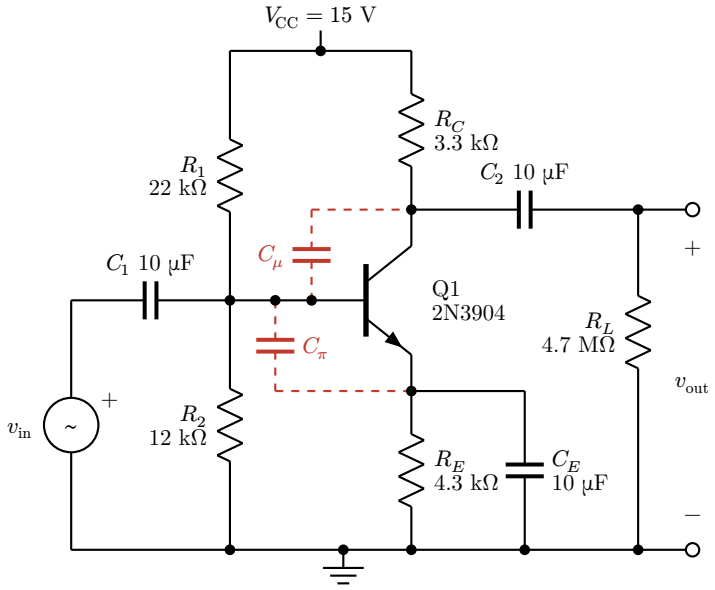


Figure 2: Common-emitter amplifier with the BJT parasitic capacitances C_π and C_μ added.

Step 1.3: Estimate their values at the bias point (2N3904 SPICE model: $C_{jc0} = 3.64$ pF, $\varphi_c = 0.75$ V, $m_c = 0.31$, $\tau_F = 301$ ps, forward-biased $C_{je} \approx 6.4$ pF).

$$C_\mu = \frac{C_{jc0}}{(1 + V_{CB}/\varphi_c)^{m_c}} = \frac{3.64\ \text{pF}}{(1 + 6.3/0.75)^{0.31}} \approx 1.8\ \text{pF}$$

$$C_\pi = g_m \tau_F + C_{je} = (40.3\ \text{mS})(301\ \text{ps}) + 6.4\ \text{pF} \approx 19\ \text{pF}$$

Check: $f_T = g_m/\big(2\pi(C_\pi + C_\mu)\big) \approx 310$ MHz, consistent with the 300 MHz datasheet value.

Task 2: Gain as $f \to 0$

Step 2.1: Capacitor impedances. $|Z_C| = 1/(\omega C) \to \infty$ as $f \to 0$, so C_1, C_2 and C_E become open circuits (C_π and C_μ are already open). Figure 3 shows the result.

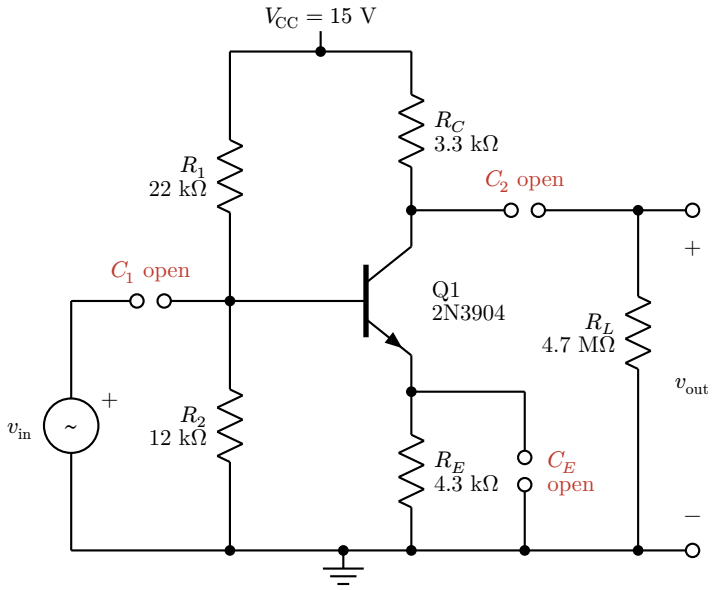


Figure 3: The amplifier as $f \to 0$: every capacitor is an open circuit.

Step 2.2: Follow the signal. With C_1 open no AC signal reaches the base; with C_2 open none reaches R_L.

Step 2.3: Intermediate region. Going down from mid-band, C_E opens first (it has the highest low-frequency pole, Step 8.6). With C_1 and C_2 still acting as shorts, R_E is unbypassed (Figure 4). The base current v_π/r_π and the collector current $g_m v_\pi$ both flow through R_E:

$$v_{\text{in}} = v_\pi + (\beta+1)\left(\frac{v_\pi}{r_\pi}\right)R_E, \quad v_{\text{out}} = -g_m v_\pi R_L'$$

$$A_v = -\frac{\beta R_L'}{r_\pi + (\beta+1)R_E} = -\frac{150 \cdot 3.15\ \text{k}}{3.72\ \text{k} + 151 \cdot 4.3\ \text{k}} = -0.72 \quad (\approx -R_L'/R_E)$$

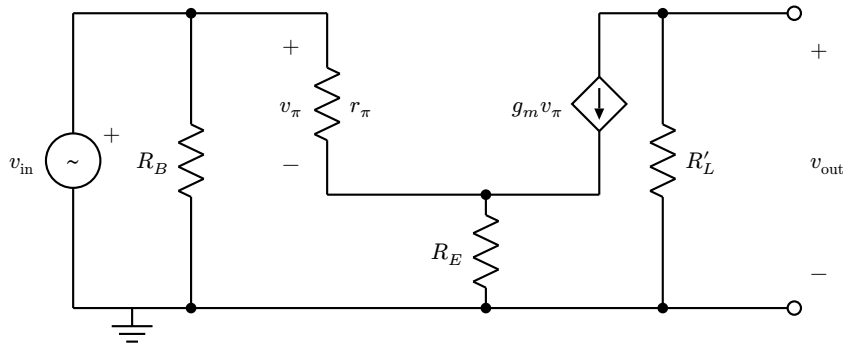


Figure 4: Small-signal circuit with C_1, C_2 shorted and C_E open (r_o is folded into R_L').

Step 2.4: Result.

$\lvert A_v \rvert \to 0$ as $f \to 0$ ($-\infty$ dB). The two zeros at $s = 0$ (from C_1 and C_2) give a +40 dB/dec slope at the lowest frequencies.

Task 3: Gain as $f \to \infty$

Step 3.1: Capacitor impedances. C_1, C_2, C_E are already shorts; now $|Z| = 1/(\omega C) \to 0$ for C_π and C_μ as well. Replace both by shorts in the hybrid-π model (Figure 5).

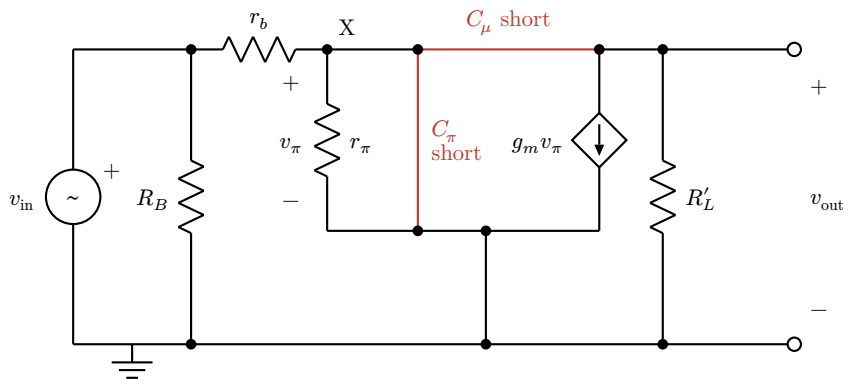


Figure 5: Small-signal circuit as $f \to \infty$: C_π and C_μ become shorts.

Step 3.2: Input side. The C_π short is in parallel with r_π, so $v_\pi = 0$ and the controlled source $g_m v_\pi = 0$.

Step 3.3: Output side. The C_μ short ties the output to the internal base, which the C_π short holds at ground, so $v_{\text{out}} = 0$.

Step 3.4: Result.

> $|A_v| \to 0$ as $f \to \infty$. With two high-frequency poles and one zero (Step 8.10), the final slope is -20 dB/dec.

Task 4: Mid-band gain

Step 4.1: Mid-band equivalent circuit. Short C_1, C_2, C_E and open C_π, C_μ (Figure 6). The emitter is at AC ground and v_{in} drives the base directly.

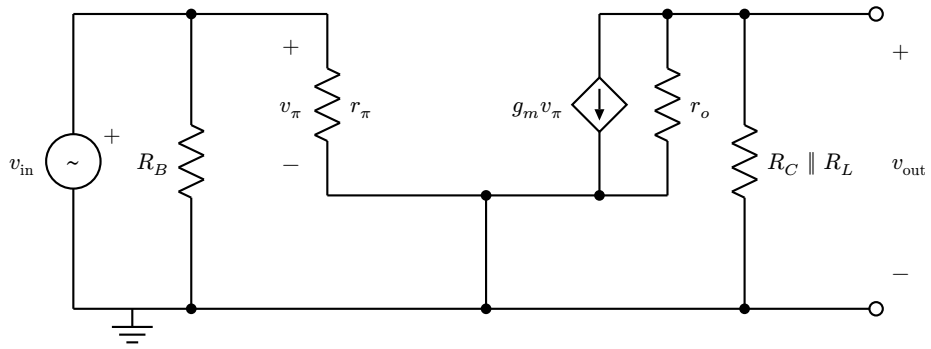


Figure 6: Mid-band small-signal circuit.

Step 4.2: Input side. r_π is connected straight across the source, so $v_\pi = v_{\text{in}}$ (with r_b included, $v_\pi = 0.997 v_{\text{in}}$).

Step 4.3: Output side. The current $g_m v_\pi$ flows through $r_o \parallel R_C \parallel R_L$:

$$v_{\text{out}} = -g_m v_\pi (R_C \parallel R_L \parallel r_o) = -g_m v_\pi R'_L$$

Step 4.4: Gain.

$$A_m = \frac{v_{\text{out}}}{v_{\text{in}}} = -g_m R'_L = -(40.3\text{ mS})(3.15\text{ k}\Omega) \approx -127, \quad 20\log_{10}127 = 42.1\text{ dB}$$

$$\boxed{A_m \approx -127\ (42.1\text{ dB});\ -133 \text{ if } r_o \text{ is neglected.}}$$

The manual quotes about -50; with R_E fully bypassed this model gives about -127.

Task 8: Poles, zeros and Bode plot

Low frequency (C_π, C_μ open)

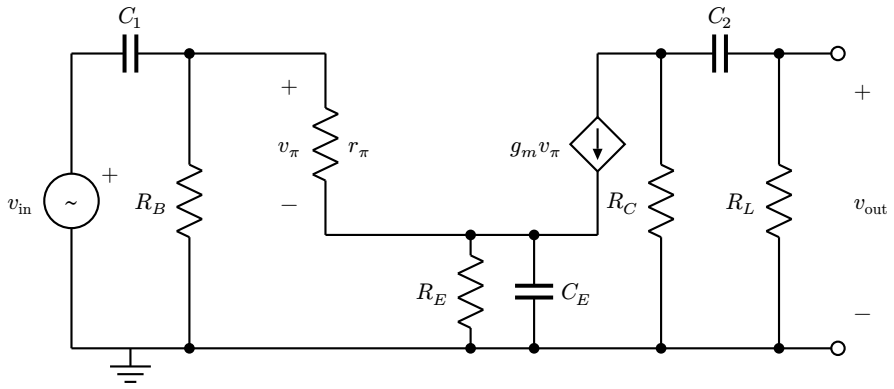


Figure 7: Low-frequency small-signal circuit (r_o neglected).

Step 8.1: Output coupling capacitor C_2 (Figure 7). The controlled source isolates C_2 from the input side; it sees $R_C \parallel r_o$ in series with R_L, giving a zero at $s = 0$ and a pole at

$$f_{p1} = \frac{1}{2\pi C_2(R_C \parallel r_o + R_L)} = \frac{1}{2\pi(10\ \mu\text{F})(4.70\ \text{M}\Omega)} = 3.38\text{ mHz}$$

Step 8.2: Emitter and base impedances.

$$Z_E = \frac{R_E}{1 + sR_E C_E}, \quad Z_b = r_\pi + (\beta + 1)Z_E$$

Step 8.3: Two gain factors.

$$\frac{v_{\text{out}}}{v_b} = -g_m R'_L \frac{r_\pi}{Z_b}, \quad \frac{v_b}{v_{\text{in}}} = \frac{sC_1 Z_{\text{in}}}{1 + sC_1 Z_{\text{in}}}, \quad Z_{\text{in}} = R_B \parallel Z_b$$

Step 8.4: Combine. Z_b cancels; with $R_0 = r_\pi + (\beta + 1)R_E = 653\ \text{k}\Omega$:

$$A_v(s) = \frac{-g_m R'_L r_\pi R_B C_1 s(1 + sR_E C_E)}{R_B(1 + sR_E C_E) + (R_0 + sr_\pi R_E C_E)(1 + sR_B C_1)}$$

Step 8.5: Zeros (numerator roots).

$$s = 0 \quad (C_1), \quad s = -\frac{1}{R_E C_E} = -23.3\text{ rad/s} \to f_{z3} = 3.70\text{ Hz} \quad (C_E)$$

Step 8.6: Poles (denominator roots). Expanding:

$$12.42s^2 + 5.120 \times 10^4 s + 6.608 \times 10^5 = 0 \quad \Rightarrow \quad s^2 + 4124s + 5.32 \times 10^4 = 0$$

$$s = -12.9\text{ rad/s} \to f_{p2} = 2.06\text{ Hz}, \quad s = -4111\text{ rad/s} \to f_{p3} = 654\text{ Hz}$$

Check, one capacitor at a time: C_E sees $R_E \parallel r_\pi/(\beta+1) = 24.5\ \Omega$, giving 650 Hz; C_1 sees $R_B \parallel R_0 = 7.67$ kΩ, giving 2.07 Hz. f_{p3} dominates, so $f_L \approx 654$ Hz.

High frequency (C_1, C_2, C_E shorted)

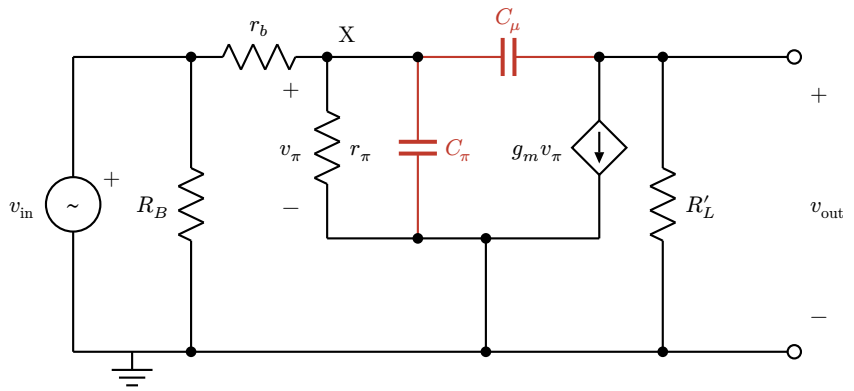


Figure 8: High-frequency small-signal circuit (hybrid-π with r_b, C_π, C_μ; r_o folded into R'_L).

Step 8.7: Thevenin equivalent at the internal base X (Figure 8, the node at the top of r_π). R_B is across the ideal source and has no effect; looking back from X:

$$v_{\text{th}} = v_{\text{in}} \frac{r_\pi}{r_\pi + r_b}, \quad R_X = r_b \parallel r_\pi = 9.97\ \Omega$$

Step 8.8: KCL at X and at the output.

$$\frac{v_{\text{th}} - v_X}{R_X} = sC_\pi v_X + sC_\mu(v_X - v_{\text{out}}), \quad sC_\mu(v_X - v_{\text{out}}) = g_m v_X + \frac{v_{\text{out}}}{R'_L}$$

Step 8.9: Solve for the transfer function.

$$A_v(s) = A_m \frac{1 - sC_\mu/g_m}{1 + b_1 s + b_2 s^2}$$

$$b_1 = R_X C_\pi + R_X(1 + g_m R'_L)C_\mu + R'_L C_\mu = 0.19 + 2.30 + 5.67 = 8.16 \text{ ns}$$

$$b_2 = R_X R'_L C_\pi C_\mu = 1.07 \times 10^{-18} \text{s}^2$$

Step 8.10: Poles and zero.

$$b_2 s^2 + b_1 s + 1 = 0 \quad \Rightarrow \quad f_{p4} = 19.8 \text{ MHz}, \quad f_{p5} = 1.19 \text{ GHz}$$

$$f_{z4} = \frac{g_m}{2\pi C_\mu} = 3.57 \text{ GHz} \quad \text{(right half-plane)}$$

Check: the dominant-pole estimate $1/(2\pi b_1) = 19.5$ MHz is close to f_{p4}, so $f_H \approx 19.8$ MHz.

Bode plot

Step 8.11: Sort the singularities and track the slope. Each zero adds +20 dB/dec and each pole −20 dB/dec; the slope is 0 in mid-band at 42.1 dB.

Table 1: Poles and zeros in frequency order and the asymptotic slope after each.

Frequency	**Singularity**	**Slope above it (dB/dec)**
< 3.38 mHz	two zeros at $s = 0$	$+40$
3.38 mHz	p_1 (C_2)	$+20$
2.06 Hz	p_2 (C_1)	0
3.70 Hz	z_3 (C_E)	$+20$
654 Hz	p_3 (C_E)	0 (mid-band, 42.1 dB)
19.8 MHz	p_4	-20
1.19 GHz	p_5	-40
3.57 GHz	z_4 (RHP)	-20

Step 8.12: Draw the asymptotes and the exact curve (Figure 9), then read off where the exact curve is 3 dB below 42.1 dB.

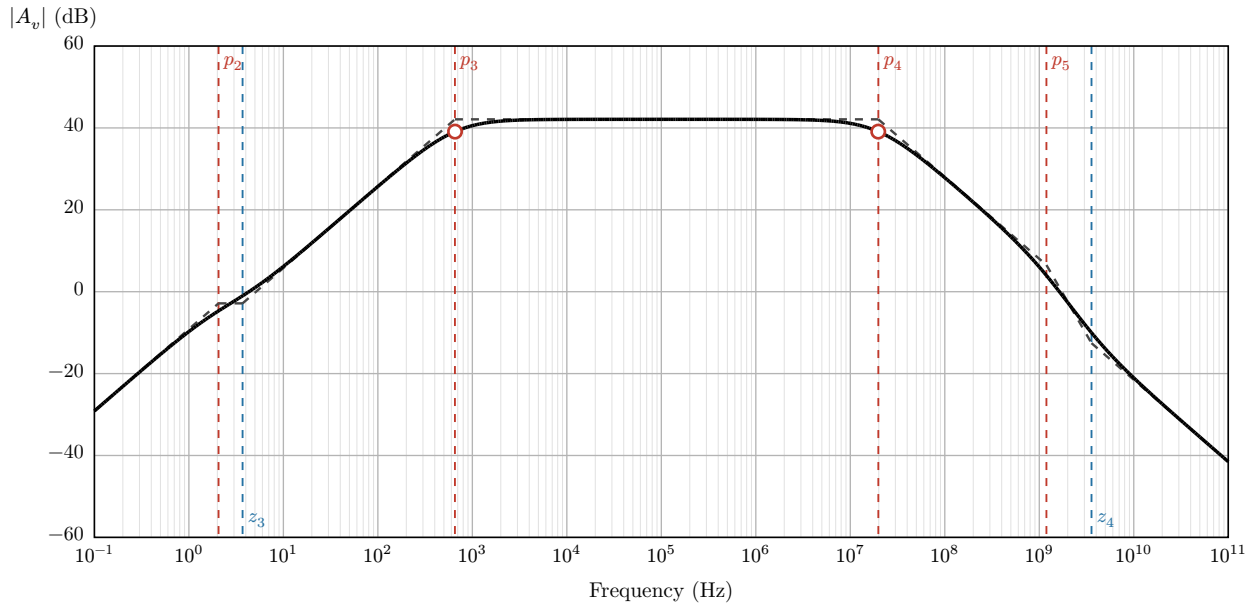


Figure 9: Voltage-gain magnitude: exact (solid) and straight-line asymptotes (dashed). Red and blue dashed lines mark poles and zeros ($p_1 = 3.4$ mHz is off scale); circles mark the 3-dB points.

$$f_L \approx 655 \text{ Hz}, \quad f_H \approx 19.7 \text{ MHz}, \quad \text{bandwidth} \approx 19.7 \text{ MHz}$$